2501-PYL1003 (Mechanics & STR) — Practice Mid-Sem Paper 5

Full Marks: 50 | Time: 2 hours | Instructor Style: Pradipta Ghosh (IIT Delhi)

Instructions

An MCQ can also be an MSQ (multiple correct options). No credit without justification/derivation.
Show all intermediate steps. Numerical answers without units lose marks.
Approximations must be justified explicitly.

Q1. A wave traveling on a stretched string has speed v that is proposed to depend on the tension F in the string, the mass per unit length μ of the string, and (possibly) the wavelength λ: $v = CF^a\mu^b\lambda^c$, with C dimensionless.

(i) Using dimensional analysis alone, show that c **cannot** be uniquely fixed to zero by dimensional arguments alone — i.e., exhibit a one-parameter family of solutions (a, b, c) that are all dimensionally consistent. **(3)**

(ii) Given the additional physical fact that wave speed on a string is independent of wavelength (non-dispersive), select the correct member of your family and state a, b. **(2)**

Q2 (MSQ). A quantity is defined as $Z = \dfrac{1}{4\pi\epsilon_0}\dfrac{q_1 q_2}{Fr^2}$, where q_1, q_2 are charges, F is the Coulomb force between them, and r is their separation. Which of the following is/are correct? **(1+1+1=3)**

a. Z is dimensionless. b. Z always evaluates to exactly 1, regardless of the values of q_1, q_2, r (as long as F is the actual Coulomb force between them). c. Z has the same dimension as ϵ_0 divided by itself, i.e., it is a pure number by construction, but its derivation nonetheless requires Coulomb's law to hold. d. If F instead denoted gravitational force between two masses (with q_1, q_2 reinterpreted as those masses), Z as defined would generally **not** be dimensionless.

Q3. A particle's velocity is given by $v(x) = \sqrt{c_1 - c_2x^2}$ for $0 \le x \le \sqrt{c_1/c_2}$, where $c_1, c_2 > 0$ are constants (this is the velocity profile of a simple harmonic oscillator, though you are not told that directly).

(i) Using $a = v\,\dfrac{dv}{dx}$, find $a(x)$ and show that it is of the form $a = -\omega^2 x$ for some constant ω; identify ω in terms of c_2. **(3)**

(ii) Without invoking the general SHM solution formula from memory, integrate $\dfrac{dx}{dt} = \sqrt{c_1 - c_2x^2}$ directly (by substitution) to find $x(t)$, given $x(0) = 0$ and $v(0) > 0$. **(3)**

Q4. A raindrop falls through a stationary cloud and grows by accretion, such that its mass increases at a rate proportional to its instantaneous cross-sectional area, which in turn is proportional to $m^{2/3}$ (since the drop is a sphere of uniform density): $\dfrac{dm}{dt} = km^{2/3}$, with $k > 0$ constant. The drop starts from rest with negligible initial mass $m_0 \to 0^+$ (idealized) and falls under gravity, with no air resistance other than the effect of accretion. Assume the accreted mass has zero momentum before joining the drop (like rain absorbing stationary droplets).

(i) Using the variable-mass equation of motion $\dfrac{d(mv)}{dt} = mg$ (no thrust term since $u_{\text{rel}} = -v$, i.e., accreted mass is at rest), and the given accretion law, show that the drop falls with **constant acceleration** g/n for some integer or simple fraction n. Find n. [Hint: try $m \propto t^3$, consistent with the accretion law solved on its own, and check self-consistency with the momentum equation.] **(6)**

Q5. A block of mass m is placed on a rough plank of mass M, which itself rests on a frictionless floor. The coefficient of friction between block and plank is μ; there is no friction between plank and floor. A constant horizontal force F is applied directly to the plank (not the block).

(i) Find the condition on F for the block and plank to move together (no relative sliding). **(3)**

(ii) For F greater than this threshold, find the acceleration of the block and of the plank separately, and find the time at which the block reaches the back edge of the plank if the block starts at a distance d from the back edge (in terms of m, M, μ, g, F, d). **(4)**

Q6. A point P has Cartesian coordinates $(x, y, z) = (5, -12, 0)$, and a point Q has Cartesian coordinates $(0, 0, 13)$.

(i) Find the spherical polar coordinates of P and of Q separately. **(3)**

(ii) Without converting back to Cartesian, use the spherical-coordinate dot product formula

$$\cos\gamma = \cos\theta_P \cos\theta_Q + \sin\theta_P \sin\theta_Q \cos(\phi_P - \phi_Q)$$

(the spherical law of cosines) to find the angle γ between the position vectors $\vec{OP}$ and $\vec{OQ}$. Verify your result using the ordinary Cartesian dot product as a check. **(4)**

Q7. A bead slides without friction on a rigid wire bent into the shape of a helix, described in cylindrical coordinates by $\rho = R$ (constant), $z = c\phi$ (constant pitch c), with $\phi(t)$ the only time-varying coordinate.

(i) Using $\vec{v} = \dot{\rho}\,\hat{\rho} + \rho\dot{\phi}\,\hat{\phi} + \dot{z}\,\hat{z}$ in cylindrical coordinates, find $\vec{v}(t)$ and the speed $|\vec{v}|$ in terms of $R, c, \dot{\phi}$. **(2)**

(ii) Using $\vec{a} = (\ddot{\rho} - \rho\dot{\phi}^2)\hat{\rho} + (\rho\ddot{\phi} + 2\dot{\rho}\dot{\phi})\hat{\phi} + \ddot{z}\,\hat{z}$, find $\vec{a}(t)$ assuming $\dot{\phi} = \omega_0$ = constant, and show that the acceleration is purely radial (i.e., has no $\hat{\phi}$ or $\hat{z}$ component) in this special case. **(3)**

Q8. Two vectors are given at the same point in space: $\vec{A} = 2\hat{r} + 3\hat{\theta} - \hat{\phi}$ and $\vec{B} = \hat{r} - \hat{\theta} + 4\hat{\phi}$, expressed in the local spherical basis at a common point $r = 5, \theta = 60^\circ, \phi = 45^\circ$.

(i) Since $\hat{r}, \hat{\theta}, \hat{\phi}$ form a right-handed orthonormal basis just like $\hat{\imath}, \hat{\jmath}, \hat{k}$, compute $\vec{A} \cdot \vec{B}$ and $\vec{A} \times \vec{B}$ directly in the spherical basis (expressing the cross product's result in terms of $\hat{r}, \hat{\theta}, \hat{\phi}$). **(4)**

(ii) Explain briefly why this computation is valid **without** first converting to Cartesian components, i.e., what property of $(\hat{r}, \hat{\theta}, \hat{\phi})$ makes this legitimate. **(1)**

Q9. A particle moves under a central force such that its orbit is a **logarithmic spiral**, $r = r_0 e^{k\phi}$, for some constants r_0, k.

(i) Using the standard substitution $u = 1/r$ and the orbit differential equation

$$\frac{d^2u}{d\phi^2} + u = -\frac{m}{L^2u^2}F(1/u)$$

find the force law $F(r)$ that produces this orbit. **(5)**

(ii) State whether this force is attractive or repulsive, and comment on how $F(r)$ scales with r (i.e., identify the power law). **(2)**

Q10. A projectile is launched from the surface of an airless planet of mass M_p and radius R_p with speed v_0 at angle θ_0 above the local horizontal. Unlike the usual flat-Earth treatment, treat gravity as a proper inverse-square central force directed toward the planet's center (i.e., this is now an orbital-mechanics problem, not projectile motion).

(i) Write down the total mechanical energy E and angular momentum L (per unit mass) of the projectile at launch, in terms of $v_0, \theta_0, R_p, M_p, G$. **(2)**

(ii) Determine the condition on v_0 (in terms of G, M_p, R_p) such that the projectile's trajectory is an ellipse with the planet's center as a focus and the launch point as the **apogee** (farthest point) of that ellipse, rather than some intermediate point of the orbit. [Hint: think about what condition on the radial velocity at launch this requires.] **(4)**

Q11. A uniform ladder of mass M and length L leans against a smooth (frictionless) vertical wall, with its base on rough ground (coefficient of static friction μ_s), making angle α with the ground. A person of mass m stands on the ladder at a distance d from the base (along the ladder).

(i) Draw the FBD of the ladder (with the person's weight included as an external force at the appropriate point) and write the three equilibrium equations (force balance in x, y, and torque balance about the base). **(3)**

(ii) Find the minimum coefficient of friction $\mu_{s,\min}$ required to prevent the ladder from slipping, in terms of M, m, L, d, α, g. **(4)**

Q12. A particle of mass m moves in a potential $U(x) = -U_0 \operatorname{sech}^2(x/a)$ (a finite, attractive potential well; $U_0, a > 0$), where $\operatorname{sech}(u) = 1/\cosh(u)$. [You do not need to know special properties of this potential beyond calculus.]

(i) Find $F(x) = -dU/dx$. **(2)**

(ii) Show that $x = 0$ is an equilibrium point and determine its stability by examining the sign of $F(x)$ for small $x > 0$ and small $x < 0$ (do **not** just compute $U''(0)$ blindly — verify your conclusion is consistent with the sign analysis). **(3)**

(iii) A particle is released from rest at $x = a$. Find its speed at $x = 0$ in terms of U_0, a, m. **(2)**

Q13. A uniform disc of mass M and radius R rolls without slipping down an incline of angle θ, starting from rest. [Moment of inertia of a disc about its central axis: $I = \frac{1}{2}MR^2$.]

(i) Using the rolling-without-slipping constraint $v_{\text{cm}} = R\omega$, and separately writing Newton's second law for translation (along the incline) and the torque equation about the center of mass, derive the linear acceleration a_{cm} of the disc's center. **(4)**

(ii) Find the minimum coefficient of static friction required for the disc to roll without slipping on this incline. **(3)**

Q14. For a system of two particles interacting via a **velocity-dependent** internal force (e.g., a magnetic-like interaction) such that $\vec{F}_{12} = q(\vec{v}_1 \times \vec{B}_{12})$ and $\vec{F}_{21} = -q(\vec{v}_2 \times \vec{B}_{21})$ for some vectors $\vec{B}_{12}, \vec{B}_{21}$ that are **not** necessarily equal and opposite in the simple Newton's-third-law sense (unlike gravity/springs):

(i) Explain, with reasoning, why linear momentum conservation for this two-particle system is **not guaranteed** in general unless a specific relation between $\vec{B}_{12}$ and $\vec{B}_{21}$ holds. State that relation. **(3)**

(ii) If instead the force is a genuine action-reaction pair, $\vec{F}_{21} = -\vec{F}_{12}$ exactly (regardless of velocities), re-derive that total momentum $\vec{p}_1 + \vec{p}_2$ is conserved, starting explicitly from Newton's second and third laws. **(2)**

Q15. A $4\ \text{kg}$ block on a frictionless horizontal surface is connected to a wall by a spring of constant $k = 100\ \text{N/m}$. The block is also subject to a constant applied horizontal force $F_0 = 20\ \text{N}$ (in the direction that stretches the spring), applied starting at $t = 0$ when the block is at rest at the spring's natural length position ($x = 0$).

(i) Find the new equilibrium position x_{eq} about which the block will oscillate. **(2)**

(ii) Write $x(t)$ explicitly for $t > 0$, and find the maximum displacement the block reaches from $x = 0$. **(3)**

(End of Paper 5 — 50 marks)